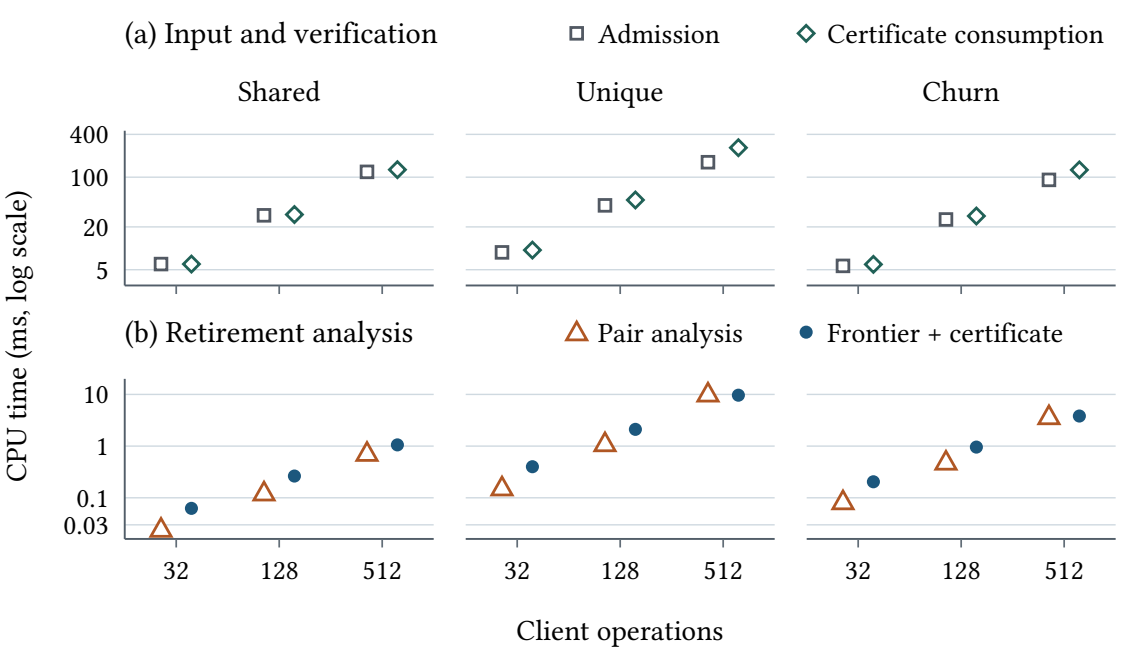

(a) Input and verification
Admission
Certificate consumption
Shared
Unique
Churn
400
100
20
5
(b) Retirement analysis
Pair analysis
Frontier + certificate
10
1
0.1
0.03
32
128
512
CPU time (ms, log scale)
Client operations